

Ultra Low Power 12-Bit SAR ADC for Wireless Sensing Applications

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Abstract—This paper presents a 12-bit SA-ADC for portable low power wireless sensor systems. The proposed SA-ADC operates for rail-to-rail input range and achieves low power consumption. Split capacitor array based DAC and a novel charge-integration based dynamic comparator are used for low power consumption of the ADC. Measured DNL and INL are -0.59/0.67 LSB and -1.2/1.33 LSB respectively. At sampling rate of 100-kS/s with 1.8-V supply, the ADC consumes only 2-μW power and achieves a SNDR of 64.42-dB, SFDR of 71.2-dB resulting in an FoM of 14-fJ/Conversion-step. The ADC core occupies an area of 0.238-mm² and is fabricated in AMS 0.35-μm CMOS technology.

Keywords—ADC; low power; successive approximation register

I. INTRODUCTION

Portable sensor systems [1] typically require an analog-to-digital converter (ADC) of 10-12 bits resolution and bandwidth of upto 100-kHz. However, wireless sensor systems operating with battery or ambient energy sources like solar, RF, vibration etc., demand extremely low power circuits and ADC is one of the important sub-blocks for signal conversion in these circuits. Typically, power consumption is expressed in energy per conversion step defined using figure of merit (FoM) given by

$$FoM = \frac{P}{2^{ENOB} F_s}$$

where, F_s is the sampling frequency, P is the power consumed and $ENOB$ is the effective number of bits. For medium resolution and sampling rates of upto few kHz, a successive approximation (SA) based ADC has been an appropriate architecture because of its low power consumption. The primary sources of power consumption in SA-ADCs are the comparator, capacitive reference digital-to-analog (DAC) array and digital control logic circuit. The power consumption of digital circuits scales with the advancement of the technology. The supply voltage reduction is the effective way of reducing power consumption since the power of digital circuits directly benefits from supply voltage reduction. Therefore, the SA-ADC in the present work is operated at 1.8v as compared to typical 3.3v for AMS 0.35μm technology. The low supply voltage makes the design of analog circuits part

more challenging. However, changes in circuit topology needs to be done to analog circuits to meet the challenges while lowering the supply voltage. Dual supply is another method of reducing power consumption at the cost of increased pins in which the analog circuits are operated at higher supply voltage and digital circuits at lower supply voltage. Instead of lowering the supply voltage, various circuit schemes [1-6] have been proposed to reduce the power consumption in ADCs by scaling of sampling rate, reducing the number of boosted switches, charge-based schemes and time-domain comparator. A SA-ADC can be implemented using fully differential, pseudo-differential and singled ended architectures. A fully-differential implementation of the ADC reduces the common-mode noise and signal distortion. A rail-to-rail operation would exclude the use of additional reference to the ADC that reduces the overall power consumption at system level.

This paper proposes a novel charge-integration based dynamic comparator for a fully-differential rail-to-rail 12-bit SA-ADC has been designed. The prototype design is fabricated in AMS 0.35μm CMOS process. The proposed ADC has FoM of 14-fJ/conversion-step for 1.8v supply and occupies an active area of 0.238-mm². The remainder of the paper is organized as follows. Section II discusses the circuit architecture of SA-ADC and implementation of various sub-blocks. Section III presents the measurement results of the proposed ADC. Finally, Section IV draws the conclusion.

II. CIRCUIT ARCHITECTURE

A. SA-ADC Architecture

The architecture of 12-bit SA-ADC is shown in Fig. 1. The fundamental building blocks of the ADC are the comparator, sample-and-hold (S/H) circuit, capacitor array, and digital control logic (also called SAR registers). A fully differential SA-ADC is chosen to suppress supply noise and reject common-mode noise. A charge-redistribution based DAC that serves as both S/H circuit and reference capacitor array is implemented. Since this ADC is fully differential, the operation of the two sides is complementary. For simplicity, only the positive side of the ADC operation is described. The operation of the ADC is as follows. In the reset phase, the charges on the MSB and LSB array of the DAC capacitor array are reset to common-mode voltage V_{cm} (shown in Fig. 5). In the sampling phase, the input signal V_{inp} , is sampled on to

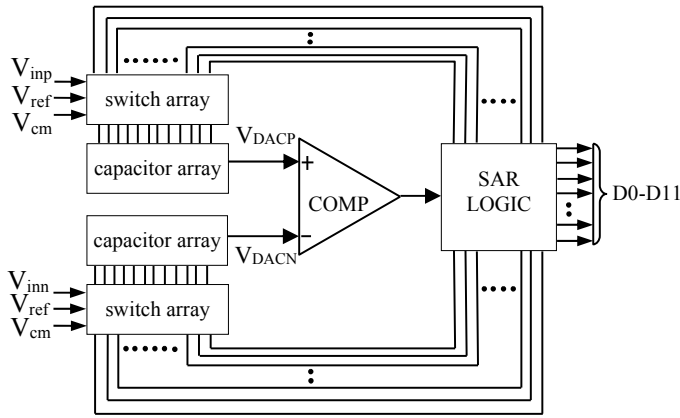


Fig. 1. Proposed SA-ADC

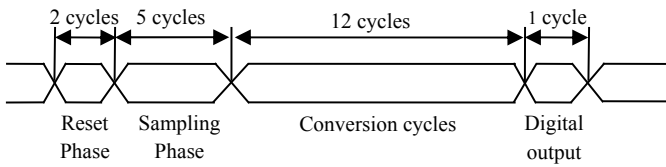


Fig. 2. Timing diagram of ADC generated by digital control logic the bottom plates of the capacitor array and top plates are still held at V_{cm} . The reset and sampling phase are overlapped to optimize the sampling and reset time of the capacitor DAC. In the conversion phase, the largest capacitor is switched to V_{ref} and the other capacitors are switched to gnd. The comparator then performs the first comparison. If V_{inp} is higher than V_{inn} , the most significant bit (MSB) is 1 otherwise, it is 0 and the largest capacitor is reconnected to ground. Then, the second largest capacitor is switched to V_{ref} . The comparator does the comparison again based on the input voltages. The ADC repeats this procedure until the least significant bit (LSB) is decided. In the last cycle, the 12-digital output bits are stored into the output register. The SAR control logic generates the required timing signals. The timing diagram for the ADC is shown in Fig. 2. The ADC is synchronous to clock running at a frequency of upto 2MHz and also, this clock goes to SAR logic and comparator. The SAR logic is implemented with D-FFs and additional digital logic using full-custom approach to optimize area and power.

B. Comparator Architecture and Operation

A novel charge-integration based dynamic comparator is shown in Fig. 3. Initially, when the clock signal clk is low, capacitor C_{sa} is set to the reference voltage (here V_{dd}) using S1 switch. Differential input signals, V_{inp} and V_{inn} are applied to M1 and M2 respectively, that act as the constant current sinks

for given input signals. When clock signal goes high, the input voltage signals are integrated for a controlled time interval ΔT . During ΔT , the voltages V_p and V_n discharges the capacitors C_{sa} , at a constant rate and the discharge rate during charge-integration depends on the input voltage applied to the current sinks. The rate of discharge amplifies the differences in the input voltages and after ΔT instant, the voltages are compared, further amplified using pre-amplifier in the dynamic

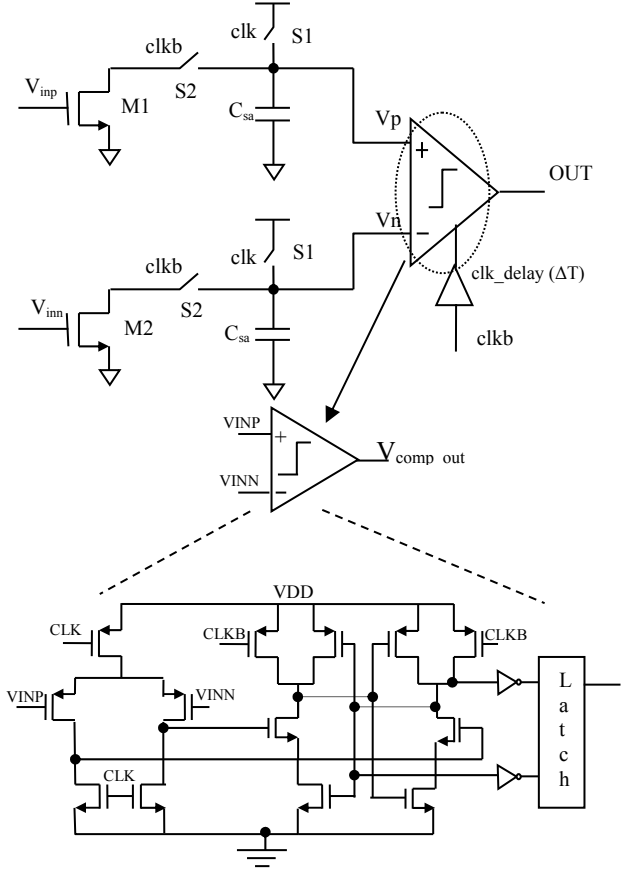


Fig. 3. Charge-integration based comparator

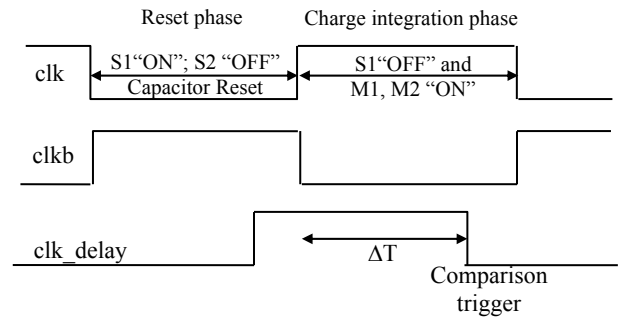


Fig. 4. Timing diagram of the comparator

comparator and finally latched to appropriate digital level. The ΔT shown in the Fig. 3, is implemented using controlled delay logic circuit that acts as the clock to the the dynamic comparator. The ΔT is obtained by delaying clk by more than half of its time period. The timing diagram of the comparator operation is shown in Fig. 4. The sizing of C_{sa} and transistors

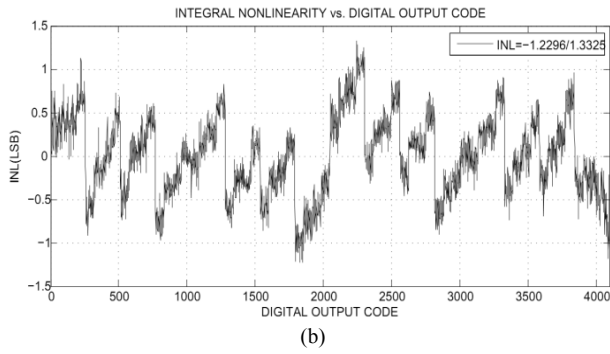


Fig. 8. Measured (a) DNL and (b) INL of ADC

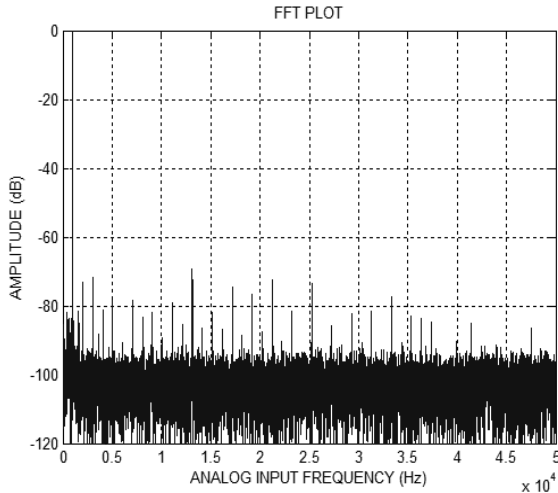


Fig. 9. Measured spectrum at 100-kS/s.

TABLE I. SUMMARY OF STATE-OF-ART 12-BIT ANALOG-TO-DIGITAL CONVERTERS

Specification	[1]	[7]	[8]	This work
Technology	0.18 μ m	0.18 μ m	0.13 μ m	0.35 μ m
Supply	1.8V	1.8V	1.2V	1.8V
Sampling	100kS/s	200kS/s	45 MS/s	100kS/s
Resolution	12-bit	12-bit	12-bit	12-bit
SNDR (dB)	65	60	70.1	64.42
SFDR (dB)	71	--	90.3	71.2
Power	25 μ W	72 μ W	3.0mW	2 μ W
FoM(fJ/step)	165	477	36.7	14

IV. CONCLUSION

This paper presents an ultra low power rail-to-rail fully differential 12-bit SA-ADC. The ADC consumes 2 μ W with 1.8V supply at a sampling rate of 100-kS/s having a FoM of 14fJ/conversion-step. The ADC achieves an ENOB of 10.41 bits. A novel charge-integrated based comparator with low offset and achieves upto 14-bit resolution having very low static power consumption is proposed. The split capacitor structure also saves capacitor area and power consumption. The proposed layout strategy for the split-capacitor DAC improves the performance of the ADC. The ADC core occupies an area of 0.238-mm² fabricated in AMS 0.35- μ m CMOS technology. The experiment results demonstrate the power and performance of the ADC.

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